

ComplaintEase – Advanced Complaint Management Platform

1. Introduction

Project Title: ComplaintEase – Advanced Complaint Management Platform

A production-ready MERN stack application for complaint registration, tracking, management, and resolution.

2. Project Overview

Purpose: To provide a centralized complaint management platform for organizations, universities, municipalities, and enterprises.

Features:

- Complaint Submission
- AI-based Auto Categorization
- Complaint Tracking Timeline
- File Uploads via Cloudinary
- Real-Time Notifications
- Comment System
- Complaint Rating
- Analytics Dashboard
- User & Department Management
- Audit Logs and Reports

3. Architecture

Frontend: React 18, Vite, Tailwind CSS, React Context, TanStack Query.

Backend: Node.js, Express.js, Socket.io.

Database: MongoDB with Mongoose schemas.

Authentication: JWT-based authentication.

Storage: Cloudinary.

Email: Resend API.

4. Setup Instructions

Prerequisites:

- Node.js v18+
- MongoDB Atlas or Local MongoDB
- Git

Installation:

1. Clone repository
2. Install server dependencies
3. Configure environment variables
4. Install client dependencies
5. Run backend and frontend servers

5. Folder Structure

Client:

components, contexts, pages, services, utils

Server:

config, controllers, models, routes, middlewares, services, utils

6. Running the Application

Frontend:

npm install

npm run dev

Backend:

npm install

npm run dev

7. API Documentation

Auth:

POST /api/auth/register

POST /api/auth/login

GET /api/auth/me

Complaints:

POST /api/complaints

GET /api/complaints

PUT /api/complaints/:id

DELETE /api/complaints/:id

Admin:

GET /api/admin/dashboard

GET /api/admin/analytics

GET /api/admin/users

8. Authentication

JWT authentication with protected routes and role-based authorization.

Roles:

- User
- Staff
- Admin

9. User Interface

Modern responsive interface with:

- Login/Register Pages
- User Dashboard
- Complaint Submission Form
- Complaint Tracking Page
- Admin Dashboard
- Analytics and Reports

10. Testing

Functional Testing

Authentication Testing

API Testing

UI Testing

Database Validation

User Acceptance Testing

11. Screenshots or Demo

Demo URL:

<https://complaint-ease-one.vercel.app/>

Include screenshots of:

- Login Page
- Dashboard
- Complaint Form
- Analytics Dashboard

12. Known Issues

- Requires internet connectivity.
- Email delivery depends on Resend configuration.
- Large file uploads may affect performance.

13. Future Enhancements

- OpenAI/Gemini Complaint Routing
- Mobile Application
- SLA-based Escalation
- SMS Notifications
- Public Complaint Portal
- Map-based Complaint Visualization